

Joint Smooth Approximation of Compact Weighted Traces and Convergence of All Variational Eigenvalues

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Status and scope

Candidate full result, likely valid, for the uniformly positive setting explicitly invoked immediately before Remark 1.3 of Vinokurov’s arXiv:2601.17896v3 [1]. The result removes the continuity and upper-boundedness assumptions from the earlier partial argument: the energy density may be rough and unbounded. It proves convergence, not merely the requested upper bound, for every eigenvalue at once.

The theorem assumes $\alpha \geq c > 0$, exactly as in the sentence preceding the open question. It does not treat a different, broader interpretation in which α may vanish. That distinction matters: the source notes that for degenerate weights the completed space $H^1(\alpha)$ can have nonunique differentials and even nonzero pure-differential elements.

Source question

Vinokurov asks in Remark 1.3, PDF page 4, whether the eigenvalues associated with an arbitrary measure inducing a compact identity $H^1(M, \alpha) \rightarrow L^2(M, \mu)$ can be approximated, or at least bounded from above, by eigenvalues associated with smooth measures [1]. The source passage, including the preceding hypothesis $\alpha \geq c > 0$, is reproduced below.

Remark 1.2. Thus, all absolutely continuous maximizing densities for $\bar{\lambda}_{k,p}$ are induced by p -harmonic maps. When $p = 2$, for λ_2 in dimension $m = 2$ and for λ_1 in dimensions $2 \leq m \leq 5$, there is a complementary result due to [KS3; KS2]: in the theorem above, μ may be any Radon measure for which the identity map $H^1(M) \rightarrow L^2(\mu)$ is compact. Note that absolutely continuous measures need not be of this kind.

Remark 1.3. The theorem above also holds for $\mu \in H^{1, \frac{m}{m-1}}(M)^*$ when $m \geq 3$ (see Remark 2.37 and Proposition 4.1). Those measures induce the compact identity map $H^1(M, \alpha) \rightarrow L^2(M, \mu)$ if $\alpha \geq c > 0$. However, we do not know whether the eigenvalues associated with an arbitrary measure for which the identity map $H^1(M, \alpha) \rightarrow L^2(M, \mu)$ is compact can be approximated (or at least bounded from above) by the eigenvalues associated with smooth measures.

Definitions

Let (M, g) be a closed connected Riemannian manifold. Let $\alpha \in L^1(M)$ satisfy $\alpha \geq c > 0$ almost everywhere, and let $\mu \neq 0$ be a finite positive Radon measure. Write

$$a_\alpha(u, v) = \int_M \alpha \langle du, dv \rangle_g dv_g, \quad b_\mu(u, v) = \int_M uv d\mu.$$

The space $H_\alpha = H^1(M, 1, \alpha)$ is the completion of smooth functions for

$$\|u\|_{H_\alpha}^2 = \|u\|_{L^2(M, v_g)}^2 + a_\alpha(u, u).$$

Uniform positivity makes the function and weak differential components unambiguous and gives a continuous injection $H_\alpha \hookrightarrow H^1(M)$. We assume that the trace

$$J : H_\alpha \longrightarrow L^2(M, \mu)$$

extends compactly from smooth functions.

The variational eigenvalues are

$$\lambda_k(\alpha, \mu) = \inf_{\substack{F \subset C^\infty(M) \\ \dim_{L^2(\mu)} F = k+1}} \sup_{0 \neq u \in F} \frac{a_\alpha(u, u)}{b_\mu(u, u)}, \quad k = 0, 1, \dots$$

Compactness of J , together with the Poincaré inequality, gives the usual discrete generalized spectrum, with $\lambda_0 = 0$.

For $2 < p < \infty$, put $q = p/(p-2)$. When $\alpha \in L^q$, Vinokurov's normalization is

$$\bar{\lambda}_{k,p}(\alpha, \mu) = \lambda_k(\alpha, \mu) \frac{\mu(M)}{\|\alpha\|_{L^q}}.$$

For $p = 2$, the source fixes $\alpha \equiv 1$.

Proof intuition

Smoothing μ alone has the wrong one-sided min-max behavior when α is rough. The key is to smooth the measures αv_g and μ by the same small diffeomorphisms.

Compactness of the limiting trace is equivalent to an Ehrling-type estimate

$$b_\mu(v, v) \leq \varepsilon a_\alpha(v, v) + C_\varepsilon \|v\|_{L^2(M, v_g)}^2.$$

Apply it to $v = u \circ \Phi_z$, then average in the smoothing parameter. The averaged mass is exactly the smoothed mass form. The averaged energy is, up to $1 + O(|z|)$, the energy of the simultaneously smoothed scalar density. Thus the entire smooth family inherits one uniform compactness estimate.

Meanwhile, the smoothed weights converge strongly in L^1 , and their energy forms converge in the Mosco sense. The uniform compactness estimate then says that strong $L^2(v_g)$ convergence plus bounded moving energy forces convergence in the moving $L^2(\mu_n)$ norms. Finite-dimensional recovery spaces give the eigenvalue limsup; limits of the first $k+1$ smooth eigenfunctions give the liminf.

Main result

Theorem 1 (joint smoothing of every compact weighted trace). *Let (M, g) be closed and connected, let $\alpha \in L^1(M)$ satisfy $\alpha \geq c > 0$ almost everywhere, and let $\mu \neq 0$ be a finite positive Radon measure such that*

$$H^1(M, 1, \alpha) \longrightarrow L^2(M, \mu)$$

is compact. There are $\alpha_n, \rho_n \in C^\infty(M)$, both strictly positive, such that, with $\mu_n = \rho_n v_g$,

$$\begin{aligned} \alpha_n &\longrightarrow \alpha && \text{in } L^1(M), \\ \mu_n &\xrightarrow{*} \mu, \quad \mu_n(M) = \mu(M), \\ \lambda_k(\alpha_n, \mu_n) &\longrightarrow \lambda_k(\alpha, \mu) && \text{for every } k \geq 0. \end{aligned}$$

If $\alpha \in L^r(M)$ for some $1 \leq r < \infty$, the construction has $\alpha_n \rightarrow \alpha$ in L^r . Consequently, for $2 < p < \infty$ and $\alpha \in L^{p/(p-2)}$, every normalized eigenvalue also converges. In the $p = 2$ case with $\alpha = 1$, one may take $\alpha_n \equiv 1$.

In particular, for $2 < p \leq m$, every pair in the uniformly positive regime of Remark 1.3 satisfies

$$\bar{\lambda}_{k,p}(\alpha, \mu) \leq \sup_{\substack{\tilde{\alpha}, \tilde{\mu} \in C^\infty(M) \\ \tilde{\alpha}, \tilde{\mu} > 0}} \bar{\lambda}_{k,p}(\tilde{\alpha}, \tilde{\mu} v_g),$$

which is the requested smooth upper bound; the theorem gives the stronger approximation statement.

Proof

1. Compactness gives an Ehrling estimate

Lemma 1. *For every $\varepsilon > 0$, there is $C_\varepsilon < \infty$ such that*

$$b_\mu(v, v) \leq \varepsilon a_\alpha(v, v) + C_\varepsilon \|v\|_{L^2(M, v_g)}^2 \quad (v \in H_\alpha). \quad (1)$$

Proof. If this failed for some ε , after normalization there would be $v_j \in H_\alpha$ with

$$b_\mu(v_j, v_j) = 1, \quad a_\alpha(v_j, v_j) \leq \varepsilon^{-1}, \quad \|v_j\|_{L^2(M, v_g)}^2 \longrightarrow 0.$$

The sequence is bounded in H_α , and any weakly convergent subsequence has weak limit zero because its $L^2(v_g)$ norm tends to zero. Compactness of J would then give $b_\mu(v_j, v_j) \rightarrow 0$, a contradiction. $\square$

2. A common intrinsic smoothing family

Choose finitely many smooth global vector fields $X_1, \dots, X_N$ whose values span $T_x M$ at every x . Such a family is obtained, for example, by embedding M in Euclidean space and projecting a basis of the ambient space onto TM . Let

$$\Phi_z = \text{Fl}_{z_N}^{X_N} \circ \dots \circ \text{Fl}_{z_1}^{X_1}, \quad z \in \mathbb{R}^N$$

for small z . Then $\Phi_0 = \text{id}$, the maps are uniformly bi-Lipschitz, and

$$(1 - C|z|)g \leq \Phi_z^* g \leq (1 + C|z|)g. \quad (2)$$

Moreover $z \mapsto \Phi_z(x)$ is a submersion near $z = 0$, uniformly in x ; these are standard consequences of smooth flows and the submersion theorem [2].

Let η_t be a nonnegative smooth density on $\mathbb{R}^N$, supported in $|z| < t$, with integral one. For a finite measure σ on M , set

$$P_t\sigma = \int_{\mathbb{R}^N} (\Phi_z)_* \sigma \eta_t(z) dz. \quad (3)$$

The submersion theorem and integration along the parameter fibers give a smooth kernel $K_t(x, y)$ such that

$$d(P_t\sigma)(y) = \left(\int_M K_t(x, y) d\sigma(x) \right) dv_g(y).$$

Thus $P_t\sigma$ has a smooth density even when σ is singular. It preserves total mass and converges weak-star to σ . If $d\sigma = f dv_g$ with $f \in L^r$, $1 \leq r < \infty$, strong continuity of pullback by a smooth family of diffeomorphisms gives

$$\frac{dP_t(fv_g)}{dv_g} \longrightarrow f \quad \text{in } L^r(M). \quad (4)$$

Define

$$\alpha_t dv_g = P_t(\alpha v_g), \quad \nu_t = P_t\mu.$$

The first density is smooth. Since $\alpha \geq c$ and the Jacobians of Φ_z equal $1 + O(t)$ uniformly,

$$\alpha_t \geq c/2 \quad (5)$$

for all sufficiently small t . To make the mass density strictly positive while preserving mass, choose $\delta_t \downarrow 0$ and put

$$\mu_t = (1 - \delta_t)\nu_t + \delta_t \frac{\mu(M)}{v_g(M)} v_g. \quad (6)$$

For smooth u , put $U_z u = u \circ \Phi_z$. The definitions give

$$b_{\nu_t}(u, u) = \int b_\mu(U_z u, U_z u) \eta_t(z) dz, \quad (7)$$

$$a_{\alpha_t}(u, u) = \int \int_M \alpha(x) |du(\Phi_z(x))|_g^2 dv_g(x) \eta_t(z) dz. \quad (8)$$

By (2), if $A_t(u) = \int a_\alpha(U_z u, U_z u) \eta_t(z) dz$, then

$$(1 - Ct)a_{\alpha_t}(u, u) \leq A_t(u) \leq (1 + Ct)a_{\alpha_t}(u, u). \quad (9)$$

Also

$$\int \|U_z u\|_{L^2(M, v_g)}^2 \eta_t(z) dz \leq (1 + Ct) \|u\|_{L^2(M, v_g)}^2. \quad (10)$$

Average (1) with $v = U_z u$ and use (7)–(10). Absorbing the harmless volume component in (6), we obtain the central estimate: for every $\varepsilon > 0$, there is C'_ε , independent of all sufficiently small t , such that

$$b_{\mu_t}(u, u) \leq \varepsilon a_{\alpha_t}(u, u) + C'_\varepsilon \|u\|_{L^2(M, v_g)}^2 \quad (u \in H^1(M)). \quad (11)$$

The estimate initially holds for smooth u and extends by density because α_t and the density of μ_t are smooth and positive.

For $p = 2$ and $\alpha = 1$, retain a_1 instead of defining a_{α_t} . Equation (2) compares $\int a_1(U_z u) \eta_t dz$ directly with $a_1(u)$, so the same estimate (11) holds with $\alpha_t = 1$.

3. Convergence of the energy and mass forms

Take any sequence $t_n \downarrow 0$, and abbreviate $a_n = a_{\alpha_{t_n}}$, $b_n = b_{\mu_{t_n}}$. By (4), $\alpha_{t_n} \rightarrow \alpha$ in L^1 . The energy forms have the following two elementary Mosco properties:

- (a) If $u_n \rightarrow u$ in $L^2(M, v_g)$ and $\sup_n a_n(u_n, u_n) < \infty$, then

$$a_\alpha(u, u) \leq \liminf_n a_n(u_n, u_n). \quad (12)$$

- (b) For every $u \in H_\alpha$, there are smooth $u_n \rightarrow u$ in $L^2(M, v_g)$ such that

$$a_n(u_n, u_n) \rightarrow a_\alpha(u, u). \quad (13)$$

For (a), (5) bounds u_n in ordinary H^1 . After taking a subsequence, $du_n \rightharpoonup du$ in L^2 , while $\sqrt{\alpha_{t_n}} \rightarrow \sqrt{\alpha}$ in L^2 . The bounded fields $\sqrt{\alpha_{t_n}} du_n$ therefore converge weakly to $\sqrt{\alpha} du$; weak lower semicontinuity proves (12). For (b), approximate u in H_α by smooth functions and choose a diagonal sequence; for each fixed smooth function, L^1 -convergence of the coefficients gives energy convergence. The same diagonal argument works simultaneously on any fixed finite-dimensional subspace and preserves all polarized energy products.

The uniform estimate upgrades this to convergence of the moving mass forms. Precisely, if

$$u_n \rightarrow u \text{ in } L^2(M, v_g), \quad \sup_n a_n(u_n, u_n) < \infty, \quad (14)$$

then

$$b_n(u_n, u_n) \rightarrow b_\mu(u, u). \quad (15)$$

To see this, choose smooth $v_j \rightarrow u$ in H_α , with their energies uniformly bounded. For fixed j , weak-star convergence gives $b_n(v_j, v_j) \rightarrow b_\mu(v_j, v_j)$. On the other hand, (11) and $a_n(u_n - v_j) \leq 2a_n(u_n) + 2a_n(v_j)$ give, for every fixed $\varepsilon > 0$,

$$\limsup_n b_n(u_n - v_j, u_n - v_j) \leq C\varepsilon + C'_\varepsilon \|u - v_j\|_{L^2(M, v_g)}^2,$$

where C is independent of j for large j . First choose ε small and then j large. The triangle inequality in $L^2(\mu_n)$, followed by the corresponding limiting argument in $L^2(\mu)$, proves (15). Polarization gives convergence of mixed mass products for finitely many such sequences.

We also need a uniform coercivity estimate. There is $C < \infty$ such that

$$\|u\|_{L^2(M, v_g)}^2 \leq C(a_n(u, u) + b_n(u, u)) \quad (16)$$

for all large n . Indeed, subtract the volume mean $\bar{u}$. The ordinary Poincaré inequality and (5) bound $\|u - \bar{u}\|_{L^2(M, v_g)}^2$ by $Ca_n(u, u)$. Estimate (11) then bounds $b_n(u - \bar{u}, u - \bar{u})$ by the same quantity. Finally,

$$\mu(M)^{1/2} |\bar{u}| = \|\bar{u}\|_{L^2(\mu_n)} \leq \|u\|_{L^2(\mu_n)} + \|u - \bar{u}\|_{L^2(\mu_n)},$$

which proves (16).

4. Two-sided min–max convergence

For the limsup, let $E \subset H_\alpha$ be the span of the first $k+1$ limiting eigenfunctions. The finite-dimensional version of the recovery construction supplies maps $R_n : E \rightarrow C^\infty(M)$ for which all energy products

converge. Equations (15) and polarization give convergence of all mass products as well. The two Gram matrices thus converge uniformly on the unit sphere of E , and the min–max principle [3] gives

$$\limsup_n \lambda_k(\alpha_{t_n}, \mu_{t_n}) \leq \lambda_k(\alpha, \mu). \quad (17)$$

For the reverse inequality, pass to a subsequence realizing the liminf. The claim is immediate if the liminf is infinite, so assume it is finite. Let $u_{n,0}, \dots, u_{n,k}$ be first eigenfunctions for the smooth pair, chosen b_n -orthonormal. Their energies are at most $\lambda_k(\alpha_{t_n}, \mu_{t_n})$. Estimate (16) bounds them in $L^2(v_g)$, and (5) bounds them in ordinary H^1 . Rellich compactness and a diagonal extraction give

$$u_{n,j} \longrightarrow u_j \quad \text{in } L^2(v_g), \quad j = 0, \dots, k.$$

Equation (15) and polarization show that the limits are b_μ -orthonormal. For every $c = (c_0, \dots, c_k)$, energy lower semicontinuity gives

$$\begin{aligned} a_\alpha \left(\sum_j c_j u_j, \sum_j c_j u_j \right) &\leq \liminf_n a_n \left(\sum_j c_j u_{n,j}, \sum_j c_j u_{n,j} \right) \\ &\leq \left(\liminf_n \lambda_k(\alpha_{t_n}, \mu_{t_n}) \right) \sum_j |c_j|^2. \end{aligned}$$

The span of the u_j 's is therefore an admissible $(k+1)$ -dimensional space in the limiting min–max problem, and

$$\lambda_k(\alpha, \mu) \leq \liminf_n \lambda_k(\alpha_{t_n}, \mu_{t_n}). \quad (18)$$

Together, (17) and (18) prove eigenvalue convergence. Property (4), preservation of $\mu(M)$, and the special $p = 2$ construction prove the remaining assertions of Theorem 1. $\square$

Verification notes

The proof was checked against four common failure modes.

1. Weak-star convergence alone gives only the wrong min–max direction. The proof never relies on it alone; (11) supplies collective compactness.
2. Translating only μ is not compatible with a rough unbounded α . Here both measures are averaged by the same maps, and (9) is a multiplicative comparison independent of pointwise upper bounds on α .
3. Strong L^1 convergence of coefficients does not by itself imply operator-norm form convergence. The proof uses the weaker but adequate Mosco liminf plus finite-dimensional recovery properties.
4. Averaging spectral inequalities can turn a rank- k exceptional space into an infinite-rank one. The argument avoids averaging spectral projectors: it averages the compactness inequality and applies min–max only after compact convergence has been proved.

The most important points for human review are the smooth-kernel consequence of the spanning-flow construction, the uniform estimate (11), and the diagonal finite-dimensional recovery used in (17).

Novelty check and limitations

A bounded search was performed before promotion. Local run indexes were searched for arXiv:2601.17896, the exact Remark 1.3 wording, compact weighted traces, smooth-measure approximation, Mosco convergence, and spectral convergence. The source of Girouard–Karpukhin–Lagacé arXiv:2004.10784 was also searched. A web pass used the exact question phrase and combinations of “joint mollification”, “weighted Dirichlet form”, “compact trace”, and “eigenvalue approximation”. It found general Mosco-convergence literature but no theorem matching Theorem 1, and no later paper explicitly answering Vinokurov’s question. Novelty confidence is therefore moderate, not definitive.

The theorem does not include weights that can vanish. If Remark 1.3 is read as asking simultaneously about arbitrary degenerate α , this packet should be reclassified as a sharp partial result. Under the uniformly positive hypothesis stated immediately before the question, it is a full approximation theorem.

Human-review recommendation

Send to an analyst working on weighted Dirichlet forms or spectral geometry. The argument is short enough to verify directly and appears to close the uniformly positive case, but the interpretation of the source’s one-argument notation $H^1(M, \alpha)$ should be confirmed with the author before a publication claim.

References

- [1] D. Vinokurov, *Eigenvalue optimization in higher dimensions and p -harmonic maps*, arXiv:2601.17896v3 (2026), especially Remark 1.3 and Section 2.6.
- [2] J. M. Lee, *Introduction to Smooth Manifolds*, 2nd ed., Graduate Texts in Mathematics 218, Springer, 2013.
- [3] E. B. Davies, *Spectral Theory and Differential Operators*, Cambridge Studies in Advanced Mathematics 42, Cambridge University Press, 1995.